

Package ‘WISP.data’

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Title Management and Analysis of WISPstation Hyperspectral Data

Version 1.0.0

Description Automate the acquisition, quality control, analysis, and visualization of spectral data collected by the 'WISPstation' fixed spectroradiometer.

License GPL (>= 3)

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<https://oggioniale.github.io/WISP.data/>

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wisp_get_reflectance_data

Get data of reflectance (level2) from WISPstation for a specific date

Description

[Stable] This function represents the main entry point for data acquisition within the WISP.data package. It connects to the official Water Insight APIs to download spectral reflectance measurements for a user-defined time interval. The function handles authentication queries the remote server and returns the retrieved data in a structured tibble format. In addition to hyperspectral reflectance data (350-900 nm), the function also retrieves the water quality parameters natively computed by the WISPstation, including: TSM (Van Der Woerd & Pasterkamp, 2008), Chla (Gons et al., 2005), Kd (Gons et al., 1998), and cpc (Simis, 2006).

Usage

```
wisp_get_reflectance_data(
  version = "1.0",
  time_from = NULL,
  time_to = NULL,
  station = NULL,
  userid = NULL,
  pwd = NULL,
  save_csv = FALSE,
  out_dir = NULL
)
```

Arguments

version	A character. It is the version of the API data. Default is "1.0".
time_from	A character. It is the date and time from which the data is requested.
time_to	A character. It is the date and time to which the data is requested.
station	A character. It is the name of the station.
userid	A character. It is the userid to access to the data service.
pwd	A character. It is the password to access to the data service.

save_csv	A logical. If TRUE, the function saves the reflectance data.
out_dir	A character. The directory where the CSV file will be saved. Default is NULL, which saves to a temporary directory (tempdir()) if save_csv = TRUE.

Value

A tibble with measurement id, measurement date, instrument name, level2_quality, set of sensor (irradiance and radiances), water quality values of TSM, Chla, Kd, and cpc as provided by instrument by default, all the reflectance values from 350 to 900 nm.

Author(s)

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Examples

```
## Not run:
# NA data
reflect_data <- wisp_get_reflectance_data(
  time_from = "2024-09-01T09:00",
  time_to = "2024-09-01T14:00",
  station = "WISPstation012",
  userid = userid,
  pwd = pwd,
  save_csv = FALSE
)

# with data
reflect_data <- wisp_get_reflectance_data(
  time_from = "2024-08-01T09:00",
  time_to = "2024-08-01T14:00",
  station = "WISPstation012",
  userid = userid,
  pwd = pwd,
  save_csv = FALSE
)

# no data for the station selected
reflect_data <- wisp_get_reflectance_data(
  time_from = "2019-06-20T09:00",
  time_to = "2019-06-20T14:00",
  station = "WISPstation012",
  userid = userid,
  pwd = pwd,
  save_csv = FALSE
)

# The two dates are not consistent
reflect_data <- wisp_get_reflectance_data(
  time_from = "2019-06-20T09:00",
```

```

    time_to = "2020-06-20T14:00",
    station = "WISPstation012",
    userid = userid,
    pwd = pwd,
    save_csv = FALSE
)

## End(Not run)

```

wisp_get_reflectance_multi_data

Get data of reflectance (level2) from WISPstation for multiple dates

Description

[Stable] This function acts as an iterative wrapper around `wisp_get_reflectance_data()`. It is specifically designed to download extended time series that exceed the limits of a single API request. The function automatically splits the user-defined time interval into daily blocks, performs sequential downloads, and aggregates all retrieved data into a single, coherent tibble.

Usage

```

wisp_get_reflectance_multi_data(
  version = "1.0",
  time_from = NULL,
  time_to = NULL,
  station = NULL,
  userid = NULL,
  pwd = NULL,
  save_csv = FALSE,
  out_dir = NULL
)

```

Arguments

version	A character. It is the version of the API data. Default is "1.0"
time_from	A character. It is the date and time from which the data is requested.
time_to	A character. It is the date and time to which the data is requested.
station	A character. It is the name of the station.
userid	A character. It is the userid to access to the data service.
pwd	A character. It is the password to access to the data service.
save_csv	A logical. If TRUE, the function saves the reflectance data.
out_dir	A character. The directory where the CSV file will be saved. Default is NULL, which saves to a temporary directory (<code>tempdir()</code>) if <code>save_csv = TRUE</code> .

Value

A tibble with measurement id, measurement date, instrument name, level2_quality, set of sensor (irradiance and radiances), water quality values of TSM (Van Der Woerd & Pasterkamp, 2008), Chla (Gons et al., 2005), Kd (Gons et al., 1998), and cpc (Simis, 2006) as provided by instrument by default, all the reflectance values from 350 to 900 nm.

Author(s)

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Examples

```
## Not run:
reflect_data <- wisp_get_reflectance_multi_data(
  time_from = "2024-04-08T09:00",
  time_to = "2024-04-10T14:00",
  station = "WISPstation012",
  userid = userid,
  pwd = pwd,
  save_csv = FALSE
)

# NA data on 2024-09-01
reflect_data <- wisp_get_reflectance_multi_data(
  time_from = "2024-08-31T09:00",
  time_to = "2024-09-02T14:00",
  station = "WISPstation012",
  userid = userid,
  pwd = pwd,
  save_csv = FALSE
)

## End(Not run)
```

wisp_plot_comparison *Comparison plot of Raw vs QC vs SR reflectance data*

Description

[Stable] This function creates an interactive side-by-side visual comparison of the different WISP-station data processing levels. Using plotly submodules, it enables the visualization of up to three aligned plots within a single interactive window: native data downloaded directly from WISPstation, processed data after QC, processed data after SR. This provides a powerful tool for visually assessing how filtering and correction algorithms modify spectral signatures, remove artifacts, and improve data quality.

Usage

```
wisp_plot_comparison(
  raw_data = NULL,
  qc_data = NULL,
  sr_data = NULL,
  raw_args = NULL,
  qc_args = NULL,
  sr_args = NULL
)
```

Arguments

<code>raw_data</code>	A tibble. The original data obtained by <code>wisp_get_reflectance_data()</code> . Default is NULL. At least one of <code>raw_data</code> , <code>qc_data</code> , or <code>sr_data</code> must be provided.
<code>qc_data</code>	A tibble. The data after <code>wisp_qc_reflectance_data()</code> operations. Default is NULL.
<code>sr_data</code>	A tibble. The data after <code>wisp_sr_reflectance_data()</code> operations. Default is NULL.
<code>raw_args</code>	A list of arguments to be passed to <code>wisp_plot_reflectance_data()</code> for the raw data plot (legend). Default is NULL.
<code>qc_args</code>	A list of arguments to be passed to <code>wisp_plot_reflectance_data()</code> for the QC data plot (legend). Default is NULL.
<code>sr_args</code>	A list of arguments to be passed to <code>wisp_plot_reflectance_data()</code> for the SR data plot (legend). Default is NULL.

Value

A plotly object comparing the spectral signatures of whichever of `raw_data`, `qc_data`, and `sr_data` were provided (1 to 3 panels, always ordered WISPstation native, then QC, then SR). The plot title reflects exactly which of them are shown, e.g. "Reflectance: QC" for a single dataset, or "Reflectance comparison: WISPstation vs SR" for two.

Author(s)

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Examples

```
if (interactive()) {
  custom_raw <- list(legend_TSM = FALSE, legend_Chla = FALSE)
  custom_qc  <- list(legend_TSM = TRUE, legend_Chla = TRUE, legend_Kd = FALSE)
  custom_sr  <- list(legend_TSM = TRUE, legend_mishra_CHL = FALSE)

  fig_comparison <- wisp_plot_comparison(
    raw_data = reflect_data,
    qc_data  = reflect_data_qc,
    sr_data  = reflect_data_sr,
```

```
    raw_args = custom_raw,  
    qc_args  = custom_qc,  
    sr_args  = custom_sr  
  )  
  print(fig_comparison)  
}
```

wisp_plot_reflectance_data

Create a plot of reflectance data

Description

[Stable] This function generates an interactive visualization of all spectral signatures contained in a dataset, based on the plotly library. It is highly flexible and can be used to display: native data downloaded directly from WISPstation, processed data after QC, processed data after SR. The function's distinctive feature is its dynamic tooltip system: when hovering over a spectral curve, users can instantly visualize the corresponding acquisition date and time, together with all associated bio-optical parameters computed for that specific measurement.

Usage

```
wisp_plot_reflectance_data(  
  data,  
  legend_TSM = TRUE,  
  legend_Chla = TRUE,  
  legend_Kd = TRUE,  
  legend_cpc = TRUE,  
  legend_scatt = FALSE,  
  legend_ratio = FALSE,  
  legend_novoa_SPM = FALSE,  
  legend_novoa_TUR = FALSE,  
  legend_jiang_TSS = FALSE,  
  legend_gons_CHL = FALSE,  
  legend_gons740_CHL = FALSE,  
  legend_NDCI = FALSE,  
  legend_mishra_CHL = FALSE,  
  legend_hue_angle = FALSE,  
  legend_dom_wavelength = FALSE,  
  legend_OWT_class = FALSE,  
  legend_OWT_score = FALSE,  
  legend_OWT_z_dist = FALSE  
)
```

Arguments

<code>data</code>	A tibble obtained by any of the functions provided by this package: <code>wisp_get_reflectance_data()</code> , or after QC and SR removal operations.
<code>legend_TSM</code>	A logical. If TRUE, the plot legend includes the TSM values. Default is TRUE.
<code>legend_Chla</code>	A logical. If TRUE, the plot legend includes the Chla values. Default is TRUE.
<code>legend_Kd</code>	A logical. If TRUE, the plot legend includes the Kd values. Default is TRUE.
<code>legend_cpc</code>	A logical. If TRUE, the plot legend includes the cpc values. Default is TRUE.
<code>legend_scatt</code>	A logical. If TRUE, the plot legend includes the scattering values. Default is FALSE.
<code>legend_ratio</code>	A logical. If TRUE, the plot legend includes the ratio values. Default is FALSE.
<code>legend_novoa_SPM</code>	A logical. If TRUE, the plot legend includes the Novoa_SPM values. Default is FALSE.
<code>legend_novoa_TUR</code>	A logical. If TRUE, the plot legend includes the Novoa_TUR values. Default is FALSE.
<code>legend_jiang_TSS</code>	A logical. If TRUE, the plot legend includes the Jiang_TSS values. Default is FALSE.
<code>legend_gons_CHL</code>	A logical. If TRUE, the plot legend includes the Gons_CHL values. Default is FALSE.
<code>legend_gons740_CHL</code>	A logical. If TRUE, the plot legend includes the Gons740_CHL values. Default is FALSE.
<code>legend_NDCI</code>	A logical. If TRUE, the plot legend includes the NDCI values. Default is FALSE.
<code>legend_mishra_CHL</code>	A logical. If TRUE, the plot legend includes the Mishra_CHL values. Default is FALSE.
<code>legend_hue_angle</code>	A logical. If TRUE, the plot legend includes the Hue_Angle values. Default is FALSE.
<code>legend_dom_wavelength</code>	A logical. If TRUE, the plot legend includes the Dominant_Wavelength values. Default is FALSE.
<code>legend_OWT_class</code>	A logical. If TRUE, the plot legend includes the OWT_class. Default is FALSE.
<code>legend_OWT_score</code>	A logical. If TRUE, the plot legend includes the OWT_score (membership grade). Default is FALSE.
<code>legend_OWT_z_dist</code>	A logical. If TRUE, the plot legend includes the OWT_z_dist (statistical distance). Default is FALSE.

Value

An interactive plotly object showing the spectral signatures of the reflectance data.

Author(s)

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Examples

```
if (interactive()) {  
  wisp_plot_reflectance_data(  
    data = reflect_data_sr,  
    legend_TSM = TRUE,  
    legend_Chla = TRUE,  
    legend_Kd = TRUE,  
    legend_cpc = TRUE,  
    legend_scatt = FALSE,  
    legend_ratio = FALSE,  
    legend_novoa_SPM = FALSE,  
    legend_novoa_TUR = FALSE,  
    legend_jiang_TSS = FALSE,  
    legend_gons_CHL = FALSE,  
    legend_gons740_CHL = FALSE,  
    legend_NDCI = FALSE,  
    legend_mishra_CHL = FALSE,  
    legend_hue_angle = FALSE,  
    legend_dom_wavelength = FALSE,  
    legend_OWT_class = FALSE,  
    legend_OWT_score = FALSE,  
    legend_OWT_z_dist = FALSE  
  )  
}
```

wisp_qc_reflectance_data

Quality Control (QC) for WISPstation reflectance data

Description

[Stable] This function performs the Quality Control (QC) process and applies different algorithms to spectral signatures. The function applies a structured sequence of QC tests (QC1 - QC6), designed to identify and remove low-quality or physically implausible spectra. In addition, the function integrates independent quality assessment metrics derived from the literature (QA and QWIP), providing robust spectral validation through established optical criteria.

Usage

```
wisp_qc_reflectance_data(
  data,
  maxPeak = 0.05,
  maxPeak_blue = 0.02,
  qa_threshold = 0.5,
  qwip_threshold = 0.2,
  calc_scatt = TRUE,
  calc_SPM = TRUE,
  calc_TUR = TRUE,
  calc_TSS = TRUE,
  calc_gons = TRUE,
  calc_gons740 = TRUE,
  calc_NDCI = TRUE,
  calc_mishra = TRUE,
  calc_dom_wave = TRUE,
  calc_OWT = TRUE,
  save_csv = FALSE,
  out_dir = NULL
)
```

Arguments

<code>data</code>	A tibble. From <code>wisp_get_reflectance_data()</code> function.
<code>maxPeak</code>	A decimal. Maximum magnitude of the spectral signatures. We recommend setting this parameter to: 0.02 for clear and oligotrophic water, 0.05 for meso- to eutrophic water, and 0.08 for hypereutrophic and highly turbid water. Default is 0.05.
<code>maxPeak_blue</code>	A decimal. Maximum magnitude 350 nm values. We recommend setting this parameter to: 0.02 (default).
<code>qa_threshold</code>	A decimal. Minimum threshold for Quality Assurance (QA). We recommend setting this parameter to: 0.5 (default). To make QC more stringent, raise the threshold.
<code>qwip_threshold</code>	A decimal. Maximum threshold for Quality Water Index Polynomial (QWIP). We recommend setting this parameter to: 0.2 (default). To make QC more stringent, decrease the threshold.
<code>calc_scatt</code>	A logical. If TRUE, the function calculates the peak due to phytoplankton scattering (690-710 nm) and the ratio of the latter to the second chlorophyll absorption peak (670-680 nm). Default is TRUE.
<code>calc_SPM</code>	A logical. If TRUE, the function calculates the SPM concentrations in according to Novoa et al. (2017). Default is TRUE.
<code>calc_TUR</code>	A logical. If TRUE, the function calculates the turbidity (FNU) in according to Novoa et al. (2017). Default is TRUE.
<code>calc_TSS</code>	A logical. If TRUE, the function calculates the TSS concentrations in according to Jiang et al. (2021). Default is TRUE.

calc_gons	A logical. If TRUE, the function calculates chlorophyll using Gons et al. (2002) algorithm (NIR ~782 nm). Default is TRUE.
calc_gons740	A logical. If TRUE, the function calculates chlorophyll using Gons et al. (2002) algorithm (NIR ~740 nm). Default is TRUE.
calc_NDCI	A logical. If TRUE, the function calculates The Normalised Difference Chlorophyll Index algorithm by Mishra and Mishra (2012). Default is TRUE.
calc_mishra	A logical. If TRUE, the function calculates chlorophyll using Mishra and Mishra (2012) algorithm. Default is TRUE.
calc_dom_wave	A logical. If TRUE, the function calculates the hue angle and the dominant wavelength (van der Woerd and Wernand, 2015, 2018). Default is TRUE.
calc_OWT	A logical. If TRUE, the function calculates the Optical Water Type classification in according to Bi and Hieronymi (2024) (OWT_class), the membership probability (OWT_score), and the average Z-score (OWT_z_dist). In addition, it adds a column with the description of the corresponding OWT class. "OWT_class" represents the optical category to which the analyzed spectral signature belongs based on Gaussian Likelihood. "OWT_score" is a value between 0 and 1 representing the fuzzy membership grade; it indicates the probability of belonging to the selected class relative to the other available classes. "OWT_z_dist" indicates the statistical distance between the observed spectrum and the class mean, weighted by its standard deviation. 0-1.5 indicate an excellent fit with the OWT reference. Values above 3 suggest that the spectral signature is an outlier or deviates significantly from the typical range of that class. Default is TRUE.
save_csv	A logical. If TRUE, the function saves the reflectance data.
out_dir	A character. The directory where the CSV file will be saved. Default is NULL, which saves to a temporary directory (tempdir()) if save_csv = TRUE.

Value

A tibble with the spectral signatures that have passed QC operation and all the extra parameters that were requested. In addition, a message containing the reason behind the elimination of each anomalous spectral signature. If parameter save_csv is TRUE, the function saves the reflectance data in a CSV file.

Author(s)

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Examples

```
# Requires a valid reflectance dataset retrieved from wisp_get_reflectance_data()
if (exists("reflect_data")) {
  reflect_data_qc <- wisp_qc_reflectance_data(
    data = reflect_data,
    maxPeak = 0.05,
    maxPeak_blue = 0.02,
```

```
    qa_threshold = 0.5,  
    qwip_threshold = 0.2,  
    calc_scatt = TRUE,  
    calc_SPM = TRUE,  
    calc_TUR = TRUE,  
    calc_TSS = TRUE,  
    calc_gons = TRUE,  
    calc_gons740 = TRUE,  
    calc_NDCI = TRUE,  
    calc_mishra = TRUE,  
    calc_dom_wave = TRUE,  
    calc_OWT = TRUE,  
    save_csv = FALSE,  
  )  
}
```

wisp_sr_reflectance_data

Sky-glint Removal (SR) for WISPstation reflectance data

Description

[Stable] This function implements a sky-glint Removal algorithm based on the methodology proposed by Jiang et al. (2020). The function operates on spectra that have already been filtered through the Quality Control process and applies a correction to obtain glint-corrected remote sensing reflectance. Following this correction, all algorithms are re-applied using the corrected reflectance, ensuring more accurate and physically consistent estimates of water constituents.

Usage

```
wisp_sr_reflectance_data(  
  qc_data,  
  calc_scatt = TRUE,  
  calc_SPM = TRUE,  
  calc_TUR = TRUE,  
  calc_TSS = TRUE,  
  calc_gons = TRUE,  
  calc_gons740 = TRUE,  
  calc_NDCI = TRUE,  
  calc_mishra = TRUE,  
  calc_dom_wave = TRUE,  
  calc_OWT = TRUE,  
  save_csv = FALSE,  
  out_dir = NULL  
)
```

Arguments

qc_data	A tibble from wisp_qc_reflectance_data() function.
calc_scatt	A logical. If TRUE, the function calculates the peak due to phytoplankton scattering (690-710 nm) and the ratio of the latter to the second chlorophyll absorption peak (670-680 nm). Default is TRUE.
calc_SPM	A logical. If TRUE, the function calculates the SPM concentrations in according to Novoa et al. (2017). Default is TRUE.
calc_TUR	A logical. If TRUE, the function calculates the turbidity (FNU) in according to Novoa et al. (2017). Default is TRUE.
calc_TSS	A logical. If TRUE, the function calculates the TSS concentrations in according to Jiang et al. (2021). Default is TRUE.
calc_gons	A logical. If TRUE, the function calculates chlorophyll using Gons et al. (2002) algorithm (NIR ~782 nm). Default is TRUE.
calc_gons740	A logical. If TRUE, the function calculates chlorophyll using Gons et al. (2002) algorithm (NIR ~740 nm). Default is TRUE.
calc_NDCI	A logical. If TRUE, the function calculates The Normalised Difference Chlorophyll Index algorithm by Mishra and Mishra (2012). Default is TRUE.
calc_mishra	A logical. If TRUE, the function calculates chlorophyll using Mishra and Mishra (2012) algorithm. Default is TRUE.
calc_dom_wave	A logical. If TRUE, the function calculates the hue angle and the dominant wavelength (van der Woerd and Wernand, 2015, 2018). Default is TRUE.
calc_OWT	A logical. If TRUE, the function calculates the Optical Water Type classification in according to Bi and Hieronymi (2024) (OWT_class), the membership probability (OWT_score), and the average Z-score (OWT_z_dist). In addition, it adds a column with the description of the corresponding OWT class. "OWT_class" represents the optical category to which the analyzed spectral signature belongs based on Gaussian Likelihood. "OWT_score" is a value between 0 and 1 representing the fuzzy membership grade; it indicates the probability of belonging to the selected class relative to the other available classes. "OWT_z_dist" indicates the statistical distance between the observed spectrum and the class mean, weighted by its standard deviation. 0-1.5 indicate an excellent fit with the OWT reference. Values above 3 suggest that the spectral signature is an outlier or deviates significantly from the typical range of that class. Default is TRUE.
save_csv	A logical. If TRUE, the function saves the reflectance data.
out_dir	A character. The directory where the CSV file will be saved. Default is NULL, which saves to a temporary directory (tempdir()) if save_csv = TRUE.

Value

A tibble with the spectral signatures after the SR operation and all the extra parameters that were requested. If parameter save_csv is TRUE, the function saves the reflectance data in a CSV file.

Author(s)

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Examples

```
# Requires a valid dataset output from wisp_qc_reflectance_data()
if (exists("reflect_data_qc")) {
  reflect_data_sr <- wisp_sr_reflectance_data(
    qc_data = reflect_data_qc,
    calc_scatt = TRUE,
    calc_SPM = TRUE,
    calc_TUR = TRUE,
    calc_TSS = TRUE,
    calc_gons = TRUE,
    calc_gons740 = TRUE,
    calc_NDCI = TRUE,
    calc_mishra = FALSE,
    calc_dom_wave = TRUE,
    calc_OWT = TRUE,
    save_csv = FALSE,
  )
}
```

wisp_trend_plot

Create a temporal trend plot of one or more parameters

Description

[Stable] This function generates interactive temporal plots for one or more parameters. It is designed to handle both high-frequency measurements within a single day and long-term time series spanning multiple months. The function can aggregate data (daily mean or median) and allows comparison of multiple parameters in the same plot if they share the same unit of measurement.

Usage

```
wisp_trend_plot(
  data,
  params = c("TSM", "Chla"),
  datetime_col = "measurement.date",
  instrument_col = "instrument.name",
  aggregate = c("none", "daily_mean", "daily_median"),
  merge_plot = FALSE,
  na.rm = TRUE,
  colors = NULL,
  title = NULL,
```

```
    return_long_df = FALSE
  )
```

Arguments

<code>data</code>	A tibble containing water quality parameters and spectral signatures.
<code>params</code>	A character vector specifying which parameters to plot. Default is <code>c("TSM", "Chla")</code> .
<code>datetime_col</code>	A character. Name of the column with datetime values. Default is <code>"measurement.date"</code> .
<code>instrument_col</code>	A character. Name of the column with instrument identifiers. Default is <code>"instrument.name"</code> .
<code>aggregate</code>	A character specifying whether to aggregate data. Options are: • <code>"none"</code>: Plots all available values (requires data for only one day). • <code>"daily_mean"</code>: Calculates and plots the daily average, including a ribbon for the Standard Deviation (SD) (requires data for multiple days). • <code>"daily_median"</code>: Calculates and plots the daily median (requires data for multiple days). Default is <code>"none"</code> .
<code>merge_plot</code>	A logical. If TRUE, parameters that share the same unit of measurement will be merged into a single plot. The function will throw an error if no common units are found among the requested params. Default is FALSE.
<code>na.rm</code>	A logical. If TRUE, NA values are ignored during aggregation. Default is TRUE.
<code>colors</code>	A character vector of colors for each parameter. Default uses viridis palette.
<code>title</code>	A character. Optional title for the plot. Default is NULL.
<code>return_long_df</code>	A logical. If TRUE, the function returns the long format dataframe used for plotting instead of the plotly object. Default is FALSE.

Value

An interactive plotly object showing the temporal trend of the selected parameters, with optional ribbons for standard deviation. If `return_long_df = TRUE`, returns a tibble in long format.

Author(s)

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Examples

```
if (interactive()) {
  # Standard plot with facets for each parameter
  fig_trend <- wisp_trend_plot(
    data      = reflect_data_sr,
    params    = c("TSM", "Chla"),
    aggregate = "none",
    merge_plot = FALSE
  )
}
```

```
)  
print(fig_trend)  
  
# Merged plot for parameters with common units  
fig_merged <- wisp_trend_plot(  
  data      = reflect_data_sr,  
  params    = c("TSM", "Novoa_SPM"),  
  aggregate = "daily_mean",  
  merge_plot = TRUE  
)  
print(fig_merged)  
}
```

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